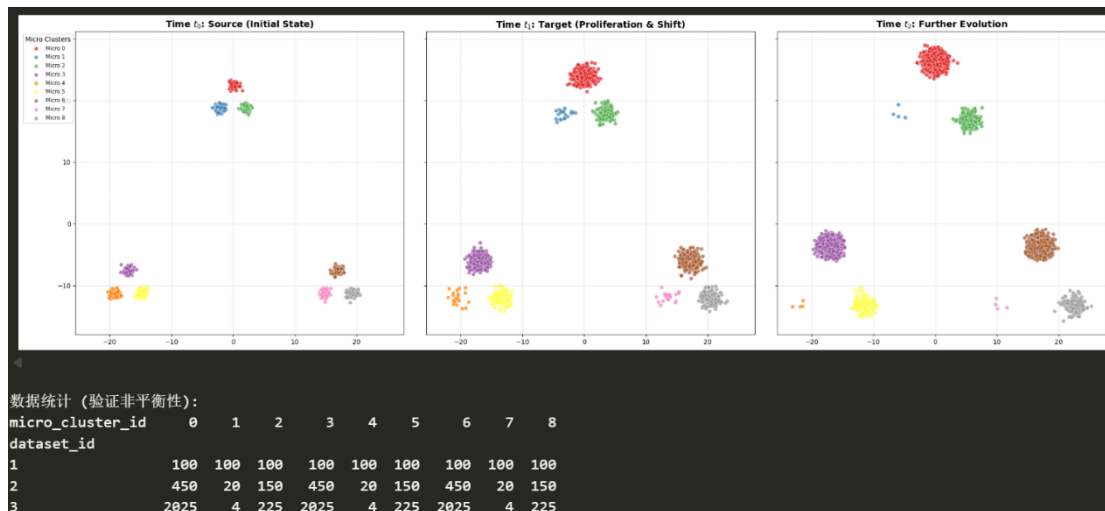
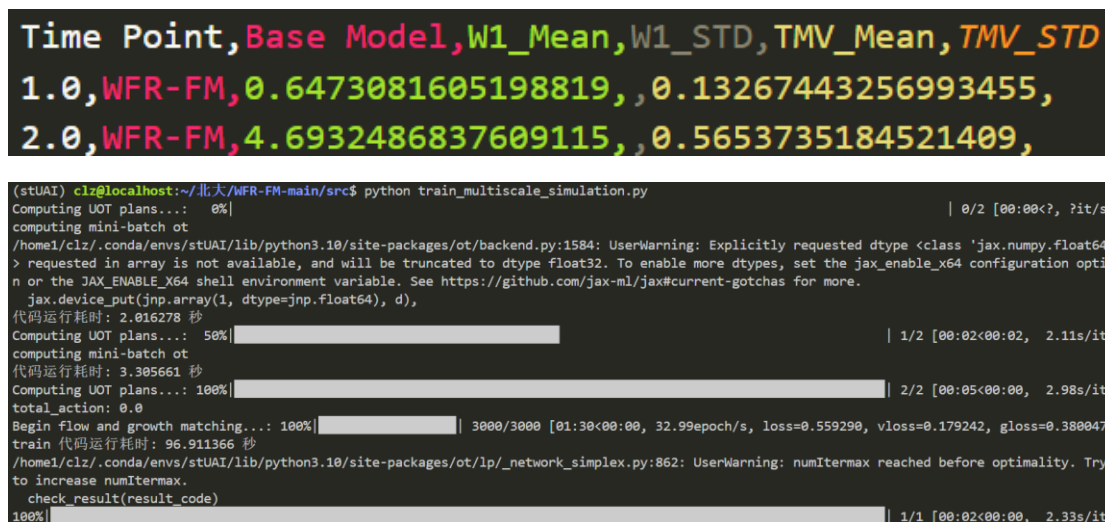
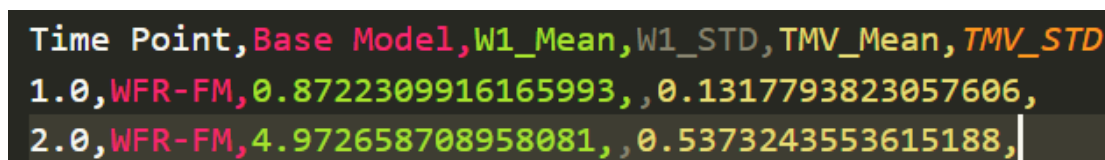




WFR-FM (mini-batch):



MS-UOT-FM:



```

(stuai) clz@localhost:~/北大/MS-UOT-FM/src$ python train_multiscale_simulation.py
Computing UOT plans...: 0% | 0/2 [00:00<?, ?it/s]
Computing Multiscale UOT plans...
Starting hierarchical WFR-OET solving (auto scales)...
Detected 2 scales: ['scale0', 'scale1']

--- Running Level 0: scale0 ---
Level 0 运行耗时: 0.027849 秒
Level 0 done. Nonzero: 3

--- Running Level 1: scale1 ---
Level 1 运行耗时: 0.024633 秒
Level 1 done. Nonzero: 9

--- Running Final Point Scale ---
Using Exact Sparse OET at point level
point ot 代码运行耗时: 0.618298 秒
Computing UOT plans...: 50% | 1/2 [00:01<00:01, 1.62s/it]
Computing Multiscale UOT plans...
Starting hierarchical WFR-OET solving (auto scales)...
Detected 2 scales: ['scale0', 'scale1']

--- Running Level 0: scale0 ---
Level 0 运行耗时: 0.016501 秒
Level 0 done. Nonzero: 3

--- Running Level 1: scale1 ---
Level 1 运行耗时: 0.017336 秒
Level 1 done. Nonzero: 8

--- Running Final Point Scale ---
Using Exact Sparse OET at point level
point ot 代码运行耗时: 1.510957 秒
Computing UOT plans...: 100% | 2/2 [00:04<00:00, 2.27s/it]
total_action: 0.0
Begin flow and growth matching...: 100% | 3000/3000 [01:35<00:00, 31.56epoch/s, loss=0.472450, vloss=0.135264, gloss=0.337186]
train 代码运行耗时: 99.613759 秒
/home1/clz/.conda/envs/stuai/lib/python3.10/site-packages/ot/lp/_network_simplex.py:862: UserWarning: numItermax reached before optimality. Try
to increase numItermax.
  check_result(result_code)
100% | 1/1 [00:02<00:00, 2.55s/it]

```

MS-UOT-FM (Independent):

```

Time Point,Base Model,W1_Mean,W1_STD,TMV_Mean,TMV_STD
1.0,WFR-FM,0.43433648682610887,,0.038004394500486406,
2.0,WFR-FM,1.1049601187122369,,0.019956585360292597,

```

```

(stuai) clz@localhost:~/北大/MS-UOT-FM/src$ python train_multiscale_simulation.py
Computing UOT plans...: 0% | 0/2 [00:00<?, ?it/s]
Computing Multiscale UOT plans...
Starting hierarchical WFR-OET solving (auto scales)...
Detected 2 scales: ['scale0', 'scale1']

--- Running Level 0: scale0 ---
Level 0 运行耗时: 0.028721 秒
Level 0 done. Nonzero: 3

--- Running Level 1: scale1 ---
Level 1 运行耗时: 0.018019 秒
Level 1 done. Nonzero: 9

--- Running Final Point Scale ---
Using Scalable Heuristic at point level
Computing UOT plans...: 50% | 1/2 [00:01<00:01, 1.12s/it]
Computing Multiscale UOT plans...
Starting hierarchical WFR-OET solving (auto scales)...
Detected 2 scales: ['scale0', 'scale1']

--- Running Level 0: scale0 ---
Level 0 运行耗时: 0.016968 秒
Level 0 done. Nonzero: 3

--- Running Level 1: scale1 ---
Level 1 运行耗时: 0.017761 秒
Level 1 done. Nonzero: 8

--- Running Final Point Scale ---
Using Scalable Heuristic at point level
Computing UOT plans...: 100% | 2/2 [00:02<00:00, 1.20s/it]
total_action: 0.0
Begin flow and growth matching...: 100% | 3000/3000 [00:41<00:00, 73.07epoch/s, loss=1.203278, vloss=0.894515, gloss=0.308763]
train 代码运行耗时: 43.455532 秒
100% | 1/1 [00:03<00:00, 3.06s/it]

```